

The Non-Injective Foundations Sub-Programme

Presentation Note 5

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Abstract

The non-injective foundations sub-programme establishes the axiomatic primitive of the Cosmochrony corpus and answers a single central question: which structures are forced once admissible non-injective projection is taken as primitive? Starting from four axioms (A1–A4) governing admissible transitions between observable states, the sub-programme derives as theorems — not postulates — the following: the non-injectivity of Π as a structural necessity of genuine emergence (ENI); the discrete Heisenberg group $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and its Weil representation V_ρ as the unique admissible fibre structure (Foundation M, HeisenbergStructure); and the absence of any independent dimensional parameter beyond c_χ (noscale). The sub-programme also provides the definitive liquidation of the preliminary vocabulary of the white paper: χ , relaxation, and iterated projection dynamics are not postulates of the axiomatic formulation but emergent consequences of A1–A4.

Keywords. non-injective projection; axiomatic primitive; Heisenberg group; Weil representation; admissibility; Born–Infeld; projection entropy; no-go theorem; emergence; foundations; presentation note.

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1 Motivation and Central Question

The non-injective foundations sub-programme answers the following question.

Central question. Is it possible to derive $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, its Weil representation, the Born–Infeld saturation constraint, and the arrow of time as consequences of a minimal set of structural axioms? More precisely: *which structures are forced once admissible non-injective projection is taken as the sole primitive?*

The sub-programme answers: all of them. Non-injectivity is not an interpretive option or a modelling choice. It is a structural necessity of any genuinely emergent description (ENI). The Heisenberg group structure is not introduced as a postulate; it is the unique algebraic form compatible with the joint constraint of irreducibility (A3) and projection locking (A4). The BI saturation constraint is the unique bounded completion of the admissible effective action. The arrow of time follows from A1+A2 alone. No background spacetime, no Hilbert space, and no quantum postulate are introduced at this level.

This note is the definitive entry point to the corpus for a reader who wants to understand *why* the specific structures of the other sub-programmes (spectral admissibility, geometry, gauge, gravity) appear, rather than merely how they are used.

2 Position in the Cosmochrony Programme

The non-injective foundations sub-programme is the axiomatic spine of the corpus. It occupies Level 1 of the three-level architecture: Level 1 (primitive), Level 2 (spectral fibre), Level 3 (physical observables).

Every other sub-programme takes one or more outputs of the foundations as inputs.

- Note 1 (Spectral Admissibility) takes $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and V_ρ , the BI bound $A_n \leq c_\chi/\sqrt{\lambda_n}$, and the parity involution as given.
- Note 2 (Emergent Geometry) takes V_ρ and the four-dimensional Carnot convergence as the starting point of the Mosco limit.
- Note 3 (Gauge Structure) takes the phase fibre and the non-injectivity of Π as the origin of gauge freedom.
- Note 4 (Spectral Gravity) takes $S_\Pi > 0$ as the structural condition that forces an effective gravitational sector.

None of these sub-programmes can be understood without the foundations. The foundations sub-programme provides the *reason* for the structures the other sub-programmes compute.

3 From the Preliminary Vocabulary to the Axiomatic Primitive

Remark 1 (Vocabulary note: liquidation of the preliminary χ framework). The white paper [3] introduced a substrate χ , a relaxation mechanism, and an iterated projection cascade $U_{t+1} = \Pi \circ \sigma(U_t)$ as the engine of dynamics. This vocabulary is superseded by the axiomatic formulation.

The axiomatic formulation retains the structural content while removing the dynamical reading: the primitive is not a substrate process, but the local structure of admissible non-injective transitions. The substrate χ is static; no relaxation mechanism is postulated at the foundational level. Observable evolution is an induced ordering of admissible projections, not a dynamics of χ itself. Relaxation and iteration appear as structural consequences of A1–A4, not as independently postulated processes.

This liquidation is final: the term “relaxation” should not appear in any description of the axiomatic framework, and χ should not be described as dynamical. In all notes of this series (Notes 1–9), the admissibility constraint replaces the earlier relaxation vocabulary.

4 Logical Chain of the Sub-Programme

The sub-programme is organised around the following derivation chain:

$$A1\text{--}A4 \implies \underbrace{\Pi \text{ non-injective, } S_\Pi > 0}_{\text{ENI, Theorem 3}} \implies \underbrace{[X, Y] \neq 0}_{\text{Foundation, Thm 5.4}} \implies \underbrace{\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})}_{\text{Foundation, Thm 5.7}} \implies \underbrace{V_\rho}_{\text{HeisStr, Thm 4.1}} .$$

Four structural implications.

1. *Non-injectivity from genuine emergence (ENI [6]).* Given surjective $\Pi : \Omega \rightarrow O$ with O not structurally isomorphic to Ω and no microstate-resolving structure beyond Π : Π is necessarily non-injective and $S_\Pi > 0$. This is a framework-independent no-go theorem, proved from surjectivity, informational completeness, and non-isomorphism alone.
2. *Non-trivial commutator from A3 (Foundation [1]).* A3 (admissibility as non-premature selection) forces the irreducibility of the admissible fibre F_n . Any decomposition $F_n = F^{(1)} \oplus F^{(2)}$ stable under admissible generators is forbidden prior to projection locking. Irreducibility forces a non-trivial commutator $[X, \sigma(X)] = Z \neq 0$ between conjugate admissible generators X and $\sigma(X)$.
3. *Heisenberg structure forced by A1–A3 and BI parity.* The non-trivial commutator $[X, Y] = Z$ generates a three-dimensional nilpotent Lie algebra. The centre is generated by Z ; its prime order q is the spectral resolution parameter. By the Stone–von Neumann uniqueness theorem, the unique irreducible unitary representation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ with central character $e^{2\pi i/q}$ is the Weil representation V_ρ .
4. *BI parity involution.* Born–Infeld parity (BornInfeld [2]) forces the minimal fibre condition $\Pi^{-1}(y) \supseteq \{\chi, -\chi\}$. Combined with A3, this closes the identification of V_ρ as the unique admissible fibre.

5 Constituent Papers

5.1 Non-injectivity as structural necessity: ENI

ENI [6] (Non-Injectivity as a Structural Necessity of Genuine Emergence) is the foundational no-go theorem.

Main theorem (Theorem 3; proved). Any surjective $\Pi : \Omega \rightarrow O$ that is informationally complete for observables and such that O is not structurally isomorphic to Ω must be non-injective with $S_\Pi > 0$. The proof uses only surjectivity, informational completeness, and the non-isomorphism condition; it is independent of any physical framework.

Categorical and informational reformulations (§8). Genuine emergence corresponds to a non-faithful functor. The Kullback–Leibler divergence of any state distribution with respect to its projection lift is strictly positive.

Corollary 6: Recursive non-injectivity (v2; proved). In any tower $\Omega_0 \xrightarrow{\Pi_1} \Omega_1 \xrightarrow{\Pi_2} \Omega_2$ of genuinely distinct emergent levels, each Π_i is non-injective with $S_{\Pi_i} > 0$, and $S_{\Pi_2 \circ \Pi_1} \geq S_{\Pi_1}$. Non-injectivity accumulates across levels: a framework with k genuinely distinct levels carries at least $k - 1$ independent non-injective projections.

Structural colour confinement (§6; structural). The tower $\Omega_0 \rightarrow \{\text{individual sectors } c_i\} \rightarrow \{\text{colour-neutral pairs and triplets}\} \rightarrow \{\text{leptons, hadrons}\}$ instantiates Corollary 6 at the colour

stratum. The colour projection $\Pi_2 : \{c_i\} \rightarrow \{\text{colour-neutral}\}$ is necessarily non-injective. The non-observability of free colour charges is therefore a structural consequence of $S_{\Pi_2} > 0$, not a dynamical accident.

Algebraic complementary route (Remark 5; structural). In frameworks whose fibre carries a Heisenberg algebra $[X, Y] = Z \neq 0$, non-trivial kernel and $S_{\Pi} > 0$ follow directly from the algebraic structure. This is complementary to Theorem 3: the abstract no-go answers *what* is forced; the algebraic route answers *why* in frameworks with Heisenberg fibres.

5.2 Axiomatic foundation: Foundation M

Foundation M [1] establishes the minimal axiomatic basis and derives from A1–A4 alone the following chain of results.

The four axioms (§2):

- A1 (Local projective admissibility): there exists a non-empty set F_n of admissible successor directions from any observable state O_{n-1} .
- A2 (Structural non-injectivity): distinct admissible directions may lead to the same resolved state; Π_n is generically non-injective.
- A3 (Non-premature selection): an admissible transition preserves the full multiplicity of open directions until projection locking; no premature selection among them is allowed.
- A4 (Projection locking): resolution occurs only when continuous BI saturation meets the discrete shell support; the transition from admissible to admitted is structurally discrete.

Derived results (proved):

- Irreversibility and the arrow of time (Proposition 4.1 and Theorem 4.2): from A1+A2 alone.
- Proto-state as physically real unresolved configuration (Proposition 5.2): from A3.
- Non-trivial commutator $[X, \sigma(X)] = Z \neq 0$ (Lemma 5.6 and its corollary): from A3 and irreducibility.
- Heisenberg group $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ as symmetry group of F_n (Theorem 5.7): from A1–A3 and BI parity.
- $F_n \cong V_\rho$ (Weil representation; stated in §5.4, detailed in HeisenbergStructure): from Theorem 5.7 and Stone–von Neumann.
- Discrete character of quantum transitions (Theorem 5.12): from A4 and O22.
- Projective incompleteness (Corollary to Theorem 5.4): every admissible Π_n has non-trivial kernel; $[X, \sigma(X)] \neq 0$ structurally forbids trivial kernel.

Three roles of Π_n (Remark 3.5; structural): admissibility filter, generator of temporal order, and revelation operator. These three roles are algebraically inseparable and all sourced in the non-commutativity.

5.3 Heisenberg structure from admissibility: HeisenbergStructure

HeisenbergStructure [5] provides the detailed proof of Foundation M Theorem 5.7, omitted from Foundation for concision. The central result (Theorem 4.1; proved):

Under A1–A3 and Born–Infeld parity, the symmetry group of F_n is isomorphic to $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for some prime q , and $F_n \simeq V_\rho$.

The proof rests on a single structural lemma (Lemma 3.1: admissible non-factorisability), followed by a chain of classification steps: non-trivial commutator, central commutator, Heisenberg algebra, prime order of the centre, Stone–von Neumann identification.

As a corollary, the Heisenberg uncertainty principle is derived as a structural property of the unresolved proto-state, not as an independent postulate.

The proof requires no input beyond A1–A3, BI parity, and classical group-classification results. It is entirely self-contained relative to Foundation M and the BornInfeld paper.

5.4 No external dimensional scale: noscale

noscale [4] establishes a structural rigidity result: the admissibility structure is closed under reparameterisations of c_χ but not under the introduction of independent dimensional parameters.

Main theorem (Theorem 3.1; proved). Any deformation of the admissibility constraints that introduces an independent dimensional parameter λ (with $[\lambda] \neq [c_\chi]^k$ for any $k \in \mathbb{Q}$) breaks at least one of: (i) BFS-consistent spectral scaling of $\sigma_{\text{pair}}(n)$; (ii) structural non-injectivity (A2) and the ENI no-go; (iii) bounded admissibility closure (A3–A4).

Corollary (proved). Dynamical fields carrying an independent dimensional coupling cannot be consistently coupled to the projection Π . In particular, a charge Q with dimensions [charge] or a cosmological constant Λ with dimensions [length] $^{-2}$ cannot enter the admissibility structure. This provides the structural backing for the exclusion of Reissner–Nordström and Schwarzschild–de Sitter geometries in Q6b.

The Level 2 statement — that every emergent scale is uniquely determined as a functional of c_χ and the cascade — is identified as an open problem (§5 of noscale).

Table 1: Summary of constituent papers. Status: P = proved; S = structural.

Paper	Central output	Status
ENI	Π non-injective $\Leftrightarrow S_\Pi > 0 \Leftrightarrow$ genuine emergence	P
ENI Cor. 6	Recursive non-injectivity; colour confinement structural	P / S
Foundation M	$A1\text{--}A4 \Rightarrow [X, Y] = Z \neq 0 \Rightarrow \text{Heis}_3 \Rightarrow V_\rho$	P
HeisenbergStructure	Detailed proof of Thm 5.7; uncertainty principle from A3	P
noscale	No independent dimensional parameter beyond c_χ	P

6 Inputs from Upstream Results

The foundations sub-programme has no upstream dependencies within the corpus. It is the axiomatic root.

The sole external input is the Born–Infeld uniqueness result of BornInfeld [2]: the BI saturation functional is the unique bounded completion of the admissible effective action. This is established independently of A1–A4 and is taken as given by the foundations sub-programme when deriving the parity involution and the identification of V_ρ .

Table 2: Principal outputs and downstream consumers.

Output	Consumer	Status
Π non-injective; $S_\Pi > 0$	Note 4 (Gravity), Q12, Q13	P
Arrow of time; irreversibility	Note 2 (Geometry), Q-series	P
$[X, Y] = Z \neq 0$ (non-commutativity)	Q1, Q2, HeisStr	P
$\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ as fibre group	All Notes 1–3	P
$V_\rho \simeq L^2(\mathbb{Z}/q\mathbb{Z})$ (Weil repr.)	Notes 1–3; Q5a–Q11	P
BI bound $A_n \leq c_\chi/\sqrt{\lambda_n}$	Note 1, O-series	P
Parity involution $c \leftrightarrow q - c$	Note 1, O18, O30	P
No independent dimensional scale beyond c_χ	Note 2 (noscale), Q6b	P
Colour confinement as structural necessity	ENI Cor. 6, O31, O32	S
Projective incompleteness	Note 3 (gauge freedom)	P

7 Outputs to Downstream Branches

8 Status of Results

8.1 Proved results

1. *ENI Theorem 3*: Π non-injective $\Leftrightarrow$ genuine emergence, proved from surjectivity, informational completeness, non-isomorphism. Framework-independent.
2. *ENI Corollary 6*: recursive non-injectivity; any tower of k distinct emergent levels carries at least $k - 1$ independent non-injective projections.
3. *Irreversibility* (Foundation Proposition 4.1): from A1+A2; the arrow of time is not postulated.
4. *Temporal ordering* (Foundation Theorem 4.2): from irreversibility and novelty.
5. *Non-factorisability Lemma 5.6* (Foundation/HeisenbergStructure): A3 forces irreducibility of F_n ; any stable decomposition prior to projection locking is forbidden.
6. *Non-trivial commutator* (Foundation Corollary to Thm 5.4): $[X, \sigma(X)] = Z \neq 0$ follows from irreducibility and the algebraic minimality of the admissible generators.
7. $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ (Foundation Theorem 5.7, HeisenbergStructure Theorem 4.1): the fibre symmetry group is uniquely the Heisenberg group of prime order; follows from the commutator structure and BI parity.
8. $F_n \cong V_\rho$ (HeisenbergStructure Theorem 4.1): Stone–von Neumann uniqueness of the irreducible representation with central character $e^{2\pi i/q}$.
9. *Uncertainty principle from A3* (HeisenbergStructure Corollary): $\Delta X \cdot \Delta Y \geq \hbar_{\text{eff}}/2$ is a structural property of the unresolved proto-state; not an independent postulate.

10. *Discrete transitions* (Foundation Theorem 5.12): from A4 and O22.
11. *Projective incompleteness*: every admissible Π_n has non-trivial kernel; no observable description is ever complete.
12. *No independent dimensional parameter* (noscale Theorem 3.1): proved for all three cases (global multiplicative, depth-dependent, configuration-dependent deformations).

8.2 Structural results

1. *Structural colour confinement* (ENI §6): colour confinement as a projective necessity of the emergence tower. The quantitative dynamics of confinement (string tension, running coupling) is not derived here; only the structural necessity of non-observability is established.
2. *Three roles of Π_n* (Foundation Remark 3.5): admissibility filter, generator of temporal order, revelation operator. These roles are algebraically inseparable and all sourced in non-commutativity.
3. *Gauge freedom as projective redundancy*: gauge freedom is not a separately postulated symmetry principle; it is the residual fibre symmetry of the non-injective projection. This is structurally stated in Foundation and detailed in Note 3.
4. *Level 2 scale determination* (noscale §5; open): that every emergent scale is uniquely determined as a functional of c_χ and the cascade is identified as an open structural problem, beyond the Level 1 rigidity proved in noscale.

8.3 Open hypotheses

1. *Born rule for general observables*. The Born rule is proved in the $SU(2)$ sector for the co-admissible spin- $\frac{3}{2}$ sector on $2I$ [7]. Extension to arbitrary observables, multipartite systems, and continuous spectra requires extending the fibre structure beyond the two-element parity case.
2. *Full Hilbert structure from discrete fibre*. The passage from the discrete $V_\rho \simeq L^2(\mathbb{Z}/q\mathbb{Z})$ to a continuous Hilbert space is handled by the Q5a continuum limit. The foundations sub-programme does not derive this passage; it delegates it to the Emergent Geometry sub-programme.
3. *Level 2 scale determination (noscale)*. That all emergent scales (masses, coupling constants, cosmological constant) are uniquely determined from c_χ is open. Only Level 1 (no independent scale can be introduced without breaking admissibility) is proved.

9 Open Deliverables

Two open deliverables bound the sub-programme.

Open deliverable 1: Born rule for general observables. The born rule is established within the $SU(2)$ sector. Its extension to arbitrary observables and to multipartite systems at the level of the foundations axioms is the primary open structural problem of this sub-programme.

Open deliverable 2: Level 2 scale determination. Proving that every emergent scale is uniquely determined from c_χ and the cascade (noscale Level 2) would make the framework fully predictive with a single dimensional free parameter. This requires a structural argument that the present programme does not yet have.

10 Conclusion

The non-injective foundations sub-programme derives the axiomatic primitive of the Cosmochrony corpus. Starting from four axioms (A1–A4) governing admissible transitions, it establishes as theorems: the structural necessity of non-injectivity (ENI); the arrow of time and irreversibility; the Heisenberg group $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and its Weil representation V_ρ as the unique admissible fibre; and the absence of any independent dimensional parameter beyond c_χ .

The white-paper vocabulary of χ , relaxation, and iterated projection is superseded. The substrate is static; the admissibility constraint replaces relaxation throughout the corpus.

These results are the inputs consumed silently by every other sub-programme. Reading the corpus without this note is possible but the origin of each structural input remains opaque. This note makes that origin explicit: every structure that the other sub-programmes compute is a theorem of A1–A4, not an independent assumption.

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